

House results: coverage and region-size trade-offs

1 SCOPE AND COMPARISON BASIS

1.1 Comparison and evidence boundaries

This report reuses the supplied House measurements rather than presenting a new experiment or full-paper reproduction. Ball is the baseline, and RCP_Ball is the column alias for RCP-Ball. The comparison uses ten observations for each method, with one observation defined as a matched seed on the same House dataset. The supplied protocol is a fixed, author-code-assisted comparison with author scales retained; seed matching was performed manually as declared in the source material.

Arithmetic was recomputed from the copied inputs, which checks the reported summaries but not data collection, implementation correctness, or scientific validity. The three quantities therefore remain on their supplied scales: coverage is a proportion, while region size and conditional coverage error use author-defined units.

2 RESULTS: A COVERAGE-REGION-SIZE TRADE-OFF

2.1 Paired numerical comparison

The central result is a three-way trade-off. Across the ten complete matched pairs, RCP-Ball has a higher mean coverage and lower mean conditional coverage error than Ball, but it also has substantially larger regions. The paired summaries are:

Quantity	RCP-Ball – Ball	Paired SE	Pairs
Coverage (proportion)	+0.004860	0.003383	10
Region size (author-defined scale)	+1.594727	0.156931	10
Conditional coverage error (author-defined scale)	−0.001349	0.000207	10

The differences are candidate minus baseline within each matched seed. The paired standard errors use the recorded convention of independent pairs; they are not confidence intervals and do not constitute significance tests.

2.2 Joint interpretation

Coverage increases from a Ball mean of 0.899052 to an RCP-Ball mean of 0.903912. Conditional coverage error decreases from 0.003184 to 0.001835 on its supplied scale. In contrast, mean region size increases from 4.625957 to 6.220684, a difference of 1.594727. The marginal ranges overlap for coverage and slightly for region size, so the table should be read as a descriptive matched comparison rather than a categorical separation of methods.

Taken together, these measurements do not support calling RCP-Ball a significant improvement or an unqualified winner. Under this fixed House comparison, the candidate shifts the two coverage-oriented summaries in the direction treated as favorable here while incurring a clear region-size cost.

Descriptive analysis of 10 supplied rows; copied data, numerical records and editable figures. Arithmetic was rechecked; data collection and scientific validity were not independently verified. Complete numerical records; full row tables are not repeated in this prose report.

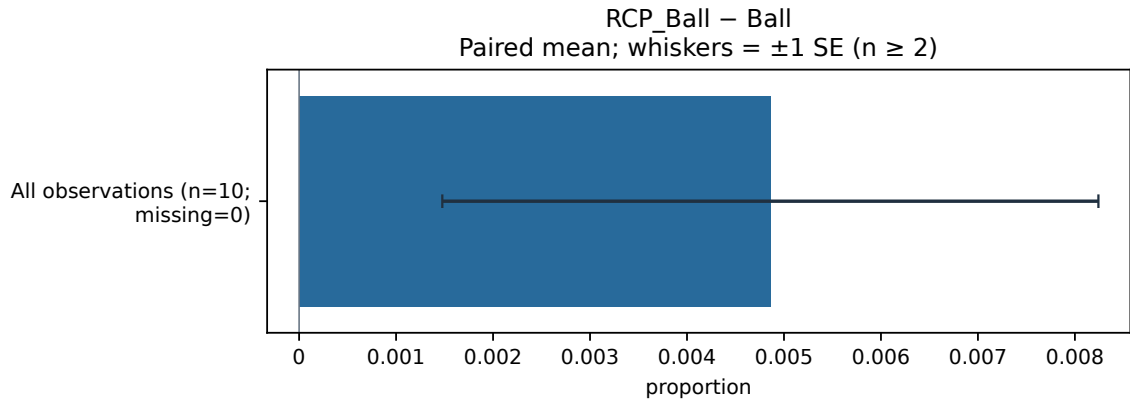


Fig. 1. RCP_Ball – Ball: matched-row mean difference (proportion); whiskers show ± 1 standard error when at least two pairs exist, assuming independent pairs, not a confidence interval. Unmatched rows are excluded. Positive does not imply improvement.

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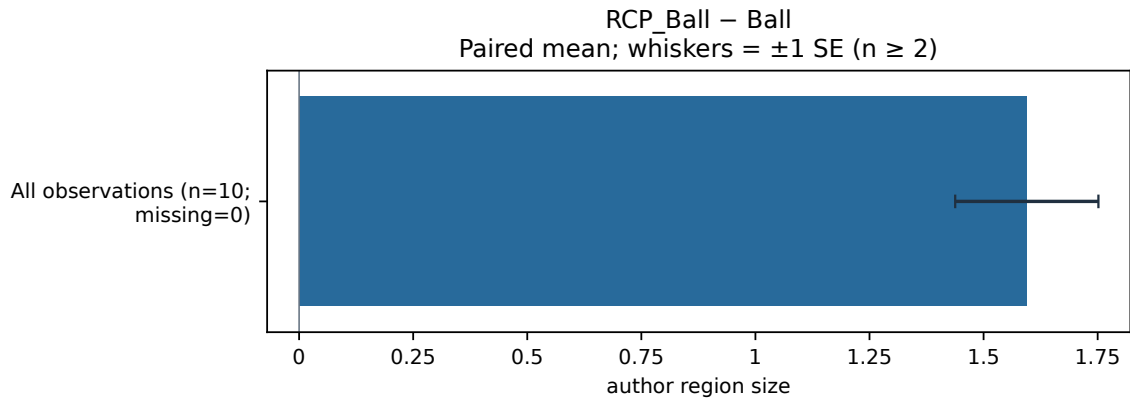


Fig. 2. RCP_Ball – Ball: matched-row mean difference (author region size); whiskers show ± 1 standard error when at least two pairs exist, assuming independent pairs, not a confidence interval. Unmatched rows are excluded. Positive does not imply improvement.

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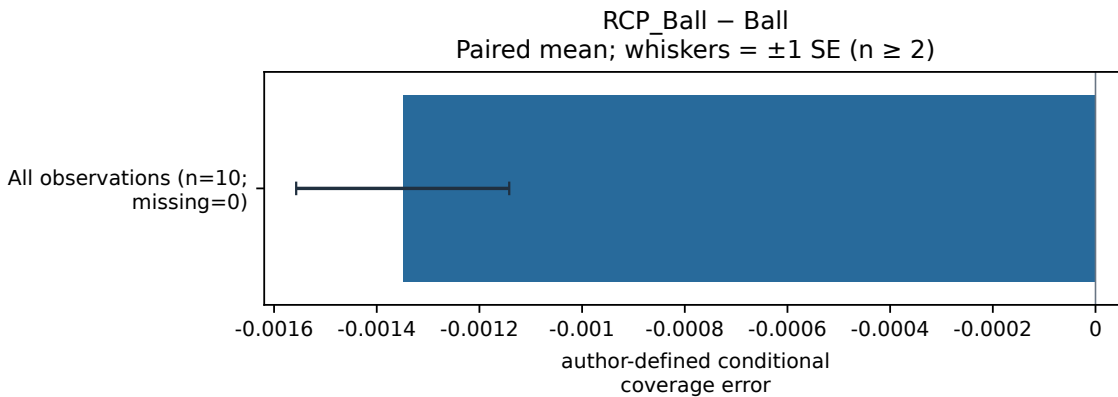


Fig. 3. RCP_Ball – Ball: matched-row mean difference (author-defined conditional coverage error); whiskers show ± 1 standard error when at least two pairs exist, assuming independent pairs, not a confidence interval. Unmatched rows are excluded. Positive does not imply improvement.

3 IMPLICATION FOR THE RESEARCH-GROUP DECISION

3.1 Decision criterion

The group should treat the choice as an application-dependent valuation problem, not as a method verdict. RCP-Ball may be attractive when higher coverage and lower conditional coverage error are more important than compact regions. Conversely, applications that place a strong cost on region size may prefer Ball's smaller regions, even though the two coverage-oriented summaries are less favorable in this sample.

The current materials do not provide a nominal target or a rule for trading these quantities against one another. Before making an adoption decision, the group should specify how much additional author-defined region size is acceptable for a given change in coverage and conditional coverage error. That rule should be agreed before interpreting a broader comparison.

3.2 Recommended interim position

RCP-Ball is promising enough to justify a targeted follow-up, but the present evidence does not justify replacing Ball by default. The interim decision should be to retain both options until the group defines the application-specific trade-off and tests whether the pattern persists beyond this supplied House comparison.

4 LIMITATIONS AND NEXT EVIDENCE

4.1 What the current evidence cannot establish

The evidence consists of ten matched seeds on one House dataset. With no significance test, the means and standard errors do not establish statistical significance, rule out seed variation, or support a conclusive superiority claim. The standard-error summaries also use an independence assumption for the pairs. The records do not establish generalization to other datasets or operating conditions, causality for the observed differences, or implementation correctness.

4.2 Focused next step

First, pre-specify the joint decision rule for coverage, region size, and conditional coverage error. Next, inspect the seed-level paired differences to determine whether the mean pattern is broadly consistent or concentrated in a few seeds. If RCP-Ball remains attractive under that rule, extend the same matched comparison to more seeds and additional datasets, preserving the pairing and applying an uncertainty analysis appropriate to the matched data. These are proposed next analyses, not completed experiments.